

Equations with Unknowns on Both Sides

AQA · KS3 Year 8

Answer each question in your workbook. Show your working.

Method

Choose the side to keep the letters Subtract the smaller letter term (both sides) Collect the numbers using inverses
Solve the simpler equation Check both sides match

Common mistakes

- Not doing the move to both sides.
- Moving the larger letter term and creating a needless negative.
- $5x - 3x = 2x$, not $8x$.
- Stopping at $ax = c$ without the final division.

Worked example

Solve $5x + 2 = 3x + 10$.

Subtract $3x$: $2x + 2 = 10$. Subtract 2 : $2x = 8$.
Divide by 2 : $x = 4$.

Check: $5(4) + 2 = 22 = 3(4) + 10$

Visual aid

Visual Example

$$2x + 3 = 11$$

$$\text{-3 both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



Balance scale with x -tiles and unit-tiles on both pans. Remove 3 x -tiles from each pan, then remove 2 units from each pan, leaving $2x = 8$ $x = 4$.

Questions

Answer each question in your workbook. Show your working.

Q1 of 8

[2] ● ○ ○ ○ ○

Solve: a) $5x = 2x + 9$ b) $7x = 3x + 20$

Q2 of 8

[2] ● ○ ○ ○ ○

Solve: a) $6x + 1 = 4x + 9$ b) $8x - 5 = 6x + 7$

Q3 of 8

[2] ● ● ○ ○ ○

Solve: a) $9x - 4 = 4x + 11$ b) $4x + 7 = x + 22$

Q4 of 8

[3] ● ● ○ ○ ○

Solve: a) $3x + 8 = 7x$ b) $2x - 5 = 5x + 4$ c) $10 - x = 4 + x$

Q5 of 8

[3] ● ● ● ○ ○

Solve: a) $5x + 3 = 2x - 9$ b) $7 - 2x = x + 1$ c) $4x - 1 = 6x - 9$

Q6 of 8

[3] ● ● ● ○ ○

Sara ($4x+2$) sweets = Tom ($6x-8$) sweets. a) Equation. b) Find x. c) How many each?

Q7 of 8

[4] ● ● ● ● ○

Plan A: $£10 + £2/\text{GB}$. Plan B: $£4 + £4/\text{GB}$. a) Equation in g. b) GB where equal. c) That cost.

Q8 of 8

[5] ● ● ● ● ●

Rod A ($5x-3$) cm = Rod B ($2x+9$) cm. a) Find x. b) Length of each. c) Rod 5 cm less than twice Rod A.